

Spy Car Game: Fun Criteria for the Build Team

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Pitch and fun pillars

The game is a one-thumb, top-down spy car brawler for iPhone: you drive an armed car up an endless highway, smash and shoot enemy cars, grab new weapons from a supply truck, and launch off ramps. Every build decision below should serve five pillars; if a feature does not serve one of them, cut it.

1. **Something exciting every 3 seconds.** A kill, a near miss, a pickup, a jump or a new threat. Dead air is the number one reason the Moonwake prototype was not fun.
2. **Contact is king.** Ramming an enemy car into a wall must be the most satisfying thing in the game. Guns are the bonus, the car itself is the main weapon.
3. **One thumb, zero friction.** Steering is the only thing the player must do. Firing is automatic. No mode switching, no holding still.
4. **Readable at a glance.** A player must know what is an enemy, a civilian, a pickup or a hazard in under a quarter second, at full speed, on a 6 inch screen.
5. **Always one more run.** Death to driving again in one tap, under 2 seconds. Every run unlocks or progresses something.

Genre reference: the classic arcade spy car shooters. The game must be original: no Spy Hunter name, car design, logo or theme music. Reuse from Moonwake: the road generator (river code becomes road code), camera, scoring and chain systems, and the water shader for boat sections.

Controls and car feel

The car must go exactly where the thumb goes, with weight but no lag; if testers ever say "the car didn't do what I wanted," controls are not done.

Control scheme (default)

- **Steer:** put a thumb anywhere on the lower 70% of the screen and slide left or right. The car's lane position follows the thumb's movement 1:1, scaled (relative drag), not a virtual joystick.

- **Fire:** automatic. Primary gun fires whenever an enemy is in front of the car's lane. The player never taps to shoot.
- **Special weapon:** one big round button, bottom right, 64pt, thumb-reachable. Shows the equipped special and ammo count. Fires on touch down, not touch up.
- **Boost / brake:** no throttle control in v1. Speed is set by the game. Boost comes from pickups and ramps only. (Throttle made Moonwake twitchy.)
- **Ram:** no button. Steer into a car to hit it. Side hits are "shunts" that push the enemy sideways.

Feel targets (starting values, tune in playtest)

Parameter	Start value	Why
Thumb to car ratio	1.4x thumb travel	Crossing the road in about 60% of screen width of thumb travel
Max lateral speed	520 pt/s	Fast enough to dodge, slow enough to aim a ram
Lateral smoothing	60 ms ease	Weight without lag; never above 90 ms
Car lean on turn	up to 10 degrees, visual only	Sells speed and turning
Forward speed	420 pt/s rising to 700 pt/s	Faster than Moonwake from second one
Auto-fire rate	6 shots/s base gun	Constant feedback, not a pea shooter
Touch dead zone	3 pt	Thumb jitter ignored, intent never ignored

Rules

- Steering is never blocked: not during jumps, not during weapon fire, not during damage.
- In the air (ramp jumps), the player can still steer at 50% authority so landings are a choice.
- Left-handed mode mirrors the special button. Sensitivity slider in settings: 0.7x to 1.6x.
- The controls must work with the phone held in one hand with the thumb low on the screen. Nothing important in the top third needs a touch.
- Game controller support (MFi and PlayStation/Xbox pads) is in scope for v1; it is cheap and press notices it.

Combat: enemy cars, ramming and civilians

Every enemy must be learnable in one encounter and beatable by both ramming and shooting, so the player always has a choice.

Enemy roster (v1 ships with 6, adds one new enemy every 60 to 90 seconds of a run)

Enemy	Behavior	Tell (warning before it acts)	Best answer	Points
Bruiser sedan	Pulls alongside and swerves into you	Brake lights flash, then it leans toward you	Ram it first, into the rail	150
Blade car	Extends tire spikes and sideswipes	Spikes glint 0.6 s before contact	Shoot from behind, or out-steer	200
Armored truck	Slow, wide, blocks lanes, immune to base gun	Heavy engine rumble, fills two lanes	Missiles, or ram it off a ramp	400
Gunner van	Fires back from behind with a mounted gun	Red laser sight line for 0.8 s	Swerve out of the sight line, then ram	250
Mine layer	Drops mines from ahead	Rear hatch opens with a click	Shoot mines for chain points, or change lane	300
Attack helicopter (from minute 3)	Strafes a lane from above	Shadow and spotlight sweep the lane first	Surface-to-air pickup, or keep moving	800

Ramming rules

- Side contact = a shunt: the enemy is knocked sideways at 1.5x the player's lateral speed. If it hits a rail, wall, another car or the road edge, it wrecks.
- Rear-ending an enemy at speed damages it; three rear hits wreck a light car.
- The player takes small damage from ramming, never a one-hit death, unless it is a head-on with a truck.
- Wrecks bounce, spin and explode, and can knock into other enemies for combo points (a "chain wreck" pays 2x).

- Pushing an enemy off a ramp edge or into oncoming traffic pays a style bonus.

Civilians

- 30 to 40% of traffic is civilian cars. Hitting or destroying one costs points and breaks the combo. It never kills the player.
- Civilians are always one color family (see Readability) and drive predictably, so weaving through them is a skill, not luck.
- Enemies can hit civilians; the player is never blamed for that.

Health instead of instant death

- The car has 3 armor pips. Damage removes one pip, with a big screen shake and a smoke trail from the car.
- Armor pickups restore one pip. Death happens at zero armor, so a single mistake never ends a run.
- No fuel meter. The pressure comes from enemies and the road, not upkeep.

Weapons and pickups

New weapons are the reward engine: the player should get a new toy every 20 to 30 seconds, and each weapon must change how they play, not just add damage.

How weapons arrive

- **Supply truck:** every 25 to 35 seconds a friendly truck (unmistakable color, open ramp at the back) appears ahead. Drive up its ramp to load a new special weapon. The load-in is a 0.8 s slow-motion beat with a big sound.
- **Crates:** enemies that are wrecked have a 15% chance to drop a crate (ammo, armor pip, or a random gun upgrade). Crates float toward the player slightly so they feel generous.
- **Ramps:** golden crates sit in the air above some ramps, only reachable by jumping.
- The player holds **one primary gun and one special at a time**. A new special replaces the old one; a duplicate refills ammo and levels it up (level 1 to 3).

Primary guns (auto-fire, swap on pickup, last until death)

Gun	What it does	Play change
Twin machine guns (default)	Straight fast shots	Baseline

Gun	What it does	Play change
Spread shot	3-way cone	Clears weaving traffic, risks civilians
Heavy cannon	Slow, big hits, knockback	Pushes enemies into walls from range
Laser	Piercing beam through several cars	Rewards lining enemies up

Specials (button, limited ammo)

Special	Ammo	What it does	Play change
Oil slick	4	Drops behind; chasing cars spin out	Answer to tailgaters and gunner vans
Smoke screen	3	Blinds cars behind, they crash into each other	Escape tool, chain wreck setup
Homing missiles	6	Lock the nearest enemy ahead	Kills trucks and helicopters
EMP pulse	2	Stalls every enemy on screen for 2 s	Panic button
Ram spikes	20 s	Wheel blades, side shunts wreck instantly	Turns the car into the weapon
Nitro	3	2 s of speed, invulnerable ram	Smash through blockades, long jumps

Weapon rules

- Every weapon has its own sound, muzzle flash, hit effect and haptic pattern. If two weapons feel the same, cut one.
- The HUD shows the special's icon and ammo as big pips on the button itself. No reading required.
- Run-start loadout: the player picks one special they have unlocked (see Progression). This gives a reason to unlock more.

Ramps, jumps and the road

Ramps are the game's signature moment: every jump should feel like a movie stunt, and the road should change often enough that no 20 seconds look the same.

Ramps and jumps

- A ramp appears every 15 to 25 seconds. It is shown 2 seconds ahead with a bright chevron sign so the player can line up.
- Air time 0.9 to 1.6 s. The camera pulls back about 15% and time slows to 70% for the first 0.3 s, then snaps back.
- The car's shadow stays on the road so the landing spot is always clear. Steering in the air is at 50%.
- Landing on an enemy car crushes it ("stomp", 3x points). Landing on a civilian is a penalty, so the player aims.
- Jumps can cross gaps: a broken bridge, a river crossing, a construction pit. Missing the ramp means taking the slower detour lane, never instant death.
- Clean landing (car straight, no damage) gives a small boost and a "Clean!" callout.

Road generator rules (reuse Moonwake's river generator)

- Road width 3 to 5 lanes, changing smoothly. Lane markings always visible so position is readable.
- Forks: the road splits for 5 to 10 seconds. One side is safer, the other has a ramp or a supply truck. A real choice every minute.
- Road edges: guard rails (wreck enemies against them), walls, open shoulders. The player scraping a rail throws sparks and grants graze points, keeping Moonwake's graze chain idea.
- Hazards on the road: barrels (explode, chain into cars), roadblocks (go around or nitro through), oil, construction cones (harmless, fun to hit).

Set pieces (one every 2 to 3 minutes, rotate in order)

1. **Boat section:** the road ends at a dock ramp; the car converts to a boat for 45 seconds on the river. Use the existing Moonwake water shader here, it is the best visual asset you have.
2. **Tunnel:** dark, headlights only, enemies revealed by their lights.
3. **Bridge chase:** narrow, rails both sides, a helicopter overhead.
4. **Downtown:** tight streets, more civilians, more ramps off parking ramps.

Biomes: coastal highway (day), desert (sunset), city (night), mountain pass (rain). The run moves through them in order so players can measure progress ("I made it to the desert").

Pacing and difficulty

The first 60 seconds are scripted, not random, and must teach everything by doing; after that the generator takes over with a difficulty curve that peaks and relaxes instead of only rising.

The first 60 seconds (scripted for every new player)

1. **0 to 3 s:** car launches off a starting ramp out of a garage. Title drops away. No text.
2. **3 to 8 s:** a single slow Bruiser pulls alongside. A ghost thumb shows "slide" once. The player rams it into the rail: first kill, huge explosion, first combo.
3. **8 to 15 s:** auto-fire kills two weak cars ahead. No instruction needed, it just happens.
4. **15 to 22 s:** civilians appear in their color. One line of text: "Spare the civilians."
5. **22 to 30 s:** first supply truck. Drive in, get homing missiles. The special button pulses until used.
6. **30 to 40 s:** armored truck that only missiles can kill. Player fires the special. Big payoff.
7. **40 to 50 s:** first ramp with a golden crate in the air.
8. **50 to 60 s:** a mixed wave of 3 enemies. Generator takes over from here.

A brand new player should not die in the first 60 seconds. Damage in this window is reduced by half.

Run structure

- Target run length: 2 to 4 minutes for a new player, 8 to 12 for a skilled one.
- Intensity works in waves: 40 to 60 s of pressure, then a 10 to 15 s breather (open road, a supply truck, a ramp). Constant pressure feels like noise.
- Every 3 minutes a boss: a heavy armored convoy or a helicopter gunship. Bosses have a health bar and 2 attack patterns with clear tells.

Difficulty curve (by minute of the run)

Minute	Enemy types live	Enemies on screen	Civilian share	Forward speed
0 to 1	Bruiser, weak cars	1 to 2	40%	420 pt/s

Minute	Enemy types live	Enemies on screen	Civilian share	Forward speed
1 to 3	+ Blade, Gunner van	2 to 3	35%	480 pt/s
3 to 5	+ Mine layer, Armored truck, helicopter	3 to 4	30%	560 pt/s
5 plus	All, in combos	4 to 5	30%	up to 700 pt/s

Fairness rules

- Every attack has a visible tell of at least 0.5 s. No off-screen deaths, ever.
- Never spawn two threats that require opposite moves at the same moment.
- Never spawn a ramp, crate or truck the player cannot physically reach.
- If the player dies, the game over screen shows what killed them, in one short line, with a 1 second replay.

Game feel: audio, haptics, camera, effects

Every player action needs a sound, a visual and a haptic response within one frame; this "juice" is what separates a fun arcade game from a tech demo, and it is not optional polish.

Impact checklist (every item must ship)

Event	Visual	Sound	Haptic
Ram / shunt	Sparks, metal crunch decal, 4 pt screen shake	Heavy metal crunch	Medium impact
Enemy wreck	Flip, fire, debris, 60 ms hit-stop	Explosion with bass tail	Heavy impact
Chain wreck	Slow-mo 0.3 s, combo text pops	Rising pitch per chain step	Double tap pattern
Gun hit	Hit flash on target, small sparks	Metallic ping	None (too frequent)
Pickup / truck load	Burst, icon flies to HUD	Bright "get" chime	Success pattern

Event	Visual	Sound	Haptic
Ramp launch	Camera pull-back, wind lines	Engine roar, wind	Light rising rumble
Landing	Dust ring, suspension squash	Thud	Medium impact
Player damage	Red edge vignette, 8 pt shake, smoke	Crunch plus alarm beep	Heavy impact
Civilian hit	Gray "-200" text, combo bar breaks	Sad horn	Light double tap
Death	1.2 s slow-mo crash, then card	Big crash, music cuts	Long heavy pattern

Audio

- Original driving music with 3 intensity layers that fade in as the action heats up. Boss music on bosses. No licensed spy themes.
- Engine sound pitch follows speed; boost is audible.
- Use AVAudioSession category `.ambient` so the game respects the silent switch and the player's own music, with a toggle for music and effects separately.

Camera

- Top-down, slight tilt (about 20 degrees) so cars read as 3D and ramps read as height.
- Camera looks ahead: the car sits at about 70% down the screen so threats ahead get the most space.
- Speed adds subtle zoom-out and motion lines. Screen shake always has a settings toggle and respects Reduce Motion.

Score feedback

- Points pop out of the thing destroyed, then fly to the score.
- Combo meter top center: multiplier grows with kills within 2.5 s of each other, up to 8x. It is the main score driver, so it must be huge and loud when it grows.

Readability and art direction

Color means role, everywhere, with no exceptions: the player must be able to play the game correctly from a blurry screenshot.

Color language (locked)

Role	Color family	Shape cue
Player car	Signature bright color plus glowing underlight	Unique silhouette, always the brightest object
Enemies	Black or gunmetal with red accents and red lights	Angular, aggressive, visible weapons
Civilians	Soft pastels (cream, light blue, mint)	Rounded, boxy family cars
Pickups and supply truck	Gold or bright green with a pulsing glow	Icons float and bob
Hazards (mines, barrels, roadblocks)	Orange and yellow hazard stripes	Blinking light
Ramps	White and gold chevrons	Arrow sign 2 s ahead

Rules

- Red is reserved for danger and enemies. Never use red for a pickup or decoration.
- Every role also has a shape cue, so color-blind players can read it. Test with a color-blind filter.
- The road is mid-value and low saturation, so cars pop off it. Effects (glow, smoke) must never hide an enemy for more than a few frames.
- Everything the player can hit or collect has a soft drop shadow on the road, so height and position are always clear.
- HUD stays in the top 15% and the special button bottom right. Nothing covers the lane ahead of the car.
- Text is short, all caps for callouts, minimum 17pt for anything the player must read during play.

Style

- Stylized 3D (low-poly with clean lighting) or a tight 2.5D. Pick one and stay consistent. Moonwake's lighting quality is the bar to hit.
- Night levels reuse the moonlight and headlight look; day levels must be just as readable.

Progression, retention and monetization

Every run must move the player forward on something, even a bad run, and monetization must never make the game feel worse to play.

Progression

- **Cash** earned every run (score converts to cash). Spent in the garage.
- **Garage:** 6 to 8 cars at launch, each with a different stat feel (heavy rammer, fast light car, armored van). Unlock with cash or by challenges.
- **Upgrades** per car: armor, ram strength, gun damage, special capacity. 5 levels each. Upgrades are felt, not just numbers.
- **Special weapon unlocks:** start with oil slick only; unlock the rest so the run-start loadout pick matters.
- **Missions:** 3 active at a time ("Wreck 5 cars with oil," "Land 3 stomps," "Reach the desert"). Completing them gives cash and progress toward a car.
- **Cosmetics:** paint, underglow color, tire trails, wreck effects. Purely visual.

Retention

- **Daily Run:** same seed for everyone each day, local-time reset with a countdown, Game Center leaderboard, streak counter.
- **Game Center:** all-time and daily leaderboards, 20 to 30 achievements.
- **Friend ghosts:** your best friend's score marker shows on the road as you approach it.
- **Return reward:** a small cash bonus for the first run each day. No guilt-trip notifications; at most one optional reminder for the Daily Run.

Monetization (pick one model before building the economy)

Option	How it works	Good for
A. Premium	\$2.99 to \$4.99 up front, no ads, no IAP	Apple featuring, reviews, simplest build
B. Free plus one unlock	Free, ads between some runs; one \$3.99 IAP removes ads and doubles cash	Reach plus a fair upsell
C. Free plus cosmetics	Free, no forced ads; sell car skins and a \$4.99 starter pack; optional rewarded ad for a continue	Long-term revenue, needs content flow

Recommendation for the build team: design the economy for option B or C, and never sell gameplay power that can't also be earned by playing. One rewarded "continue from last checkpoint" per run is fine; forced video ads mid-run are not.

Open question: which monetization model does Jack want? The economy tuning depends on it.

iOS technical requirements

Build it native (Swift with SpriteKit, SceneKit or Metal, or Unity if the team prefers), not a wrapped web page; the controls, haptics and gesture fixes below are only possible natively.

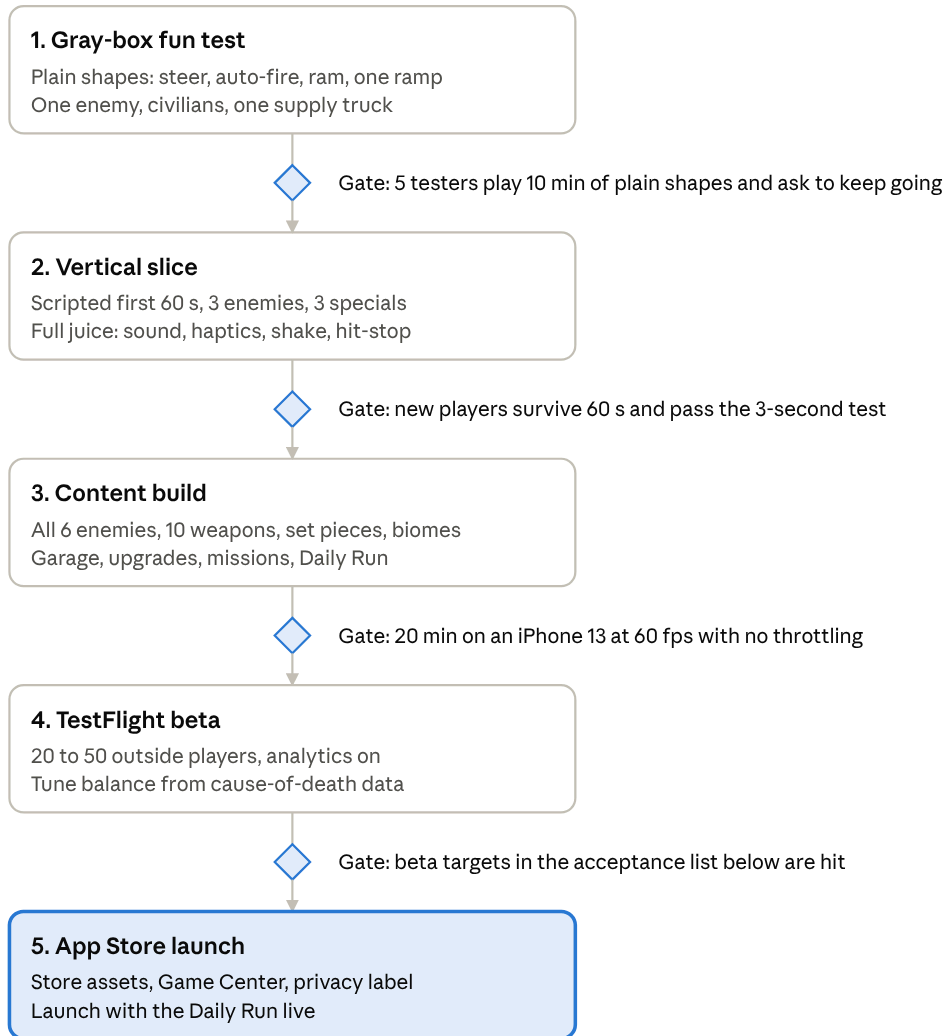
- **Frame rate:** locked 60 fps on iPhone 12 and newer; 120 fps option on ProMotion devices. All motion and timers use delta time, never per-frame steps.
- **Thermals and battery:** 20 minutes of play without visible throttling on an iPhone 13. Dynamic resolution scaling if frame time spikes. Low Power Mode drops to 30 fps and fewer effects.
- **Screen sizes:** portrait only. Layout adapts from iPhone SE to Pro Max and iPad (wider road view, no black bars). Respect the safe areas and the Dynamic Island.
- **System gestures:** defer screen edge gestures (`preferredScreenEdgesDeferringSystemGestures = .all`) and auto-hide the home indicator, so a bottom swipe does not quit the game.
- **Haptics:** Core Haptics patterns per the Game feel table, with a settings toggle.
- **Audio:** `.ambient` session, handles interruptions (calls, Siri) by pausing.

- **Pause and resume:** auto-pause when the app goes to the background; resume with a 3-2-1 countdown.
- **Load time:** cold start to driving in under 5 seconds. Death to driving again in under 2 seconds.
- **Accessibility:** Reduce Motion (less shake and flash), color-blind safe roles, adjustable control sensitivity, text at least 17pt in play, VoiceOver on all menus.
- **Offline:** fully playable offline. Leaderboards sync when back online.
- **Analytics (privacy-light):** run length, cause of death, first-session drop-off point, weapon pick rates. Declared in the App Store privacy label.
- **Store readiness:** app icon set, 6.9" and 6.5" screenshots, a 30 second preview video, age rating questionnaire, Game Center entitlement, TestFlight external beta.

Fun tests, acceptance criteria and milestones

The game is done when real players pass these tests, not when the feature list is complete; the build moves in five phases, and each gate must be passed before the next phase starts.

Nothing moves to the next phase until its gate is passed



build milestones · 5 phases, 4 gates

Phase 1 matters most: if the plain-shapes version is not fun, art and content will not fix it, so change the mechanics before moving on.

Fun tests (run with 5 or more people who have not seen the game)

- ☐ **3-second test:** record a play session; no stretch longer than 3 seconds without a kill, near miss, pickup, jump or new threat.
- ☐ **No-instructions test:** a new player gets their first kill in under 10 seconds without being told how.
- ☐ **Control test:** zero testers say the car did not go where they wanted.
- ☐ **Readability test:** shown a blurred screenshot, testers correctly name enemy, civilian, pickup and hazard 9 times out of 10.

- ☐ **Death test:** every tester can say what killed them, and 4 of 5 say it was their fault.
- ☐ **One more run test:** after dying, 4 of 5 testers restart without being asked.
- ☐ **Show-a-friend test:** testers describe a moment they want to show someone (a jump, a chain wreck).

Beta acceptance targets (TestFlight)

- ☐ Day-1 retention 35% or higher.
- ☐ Median session length 6 minutes or more.
- ☐ Average 3 or more runs per session.
- ☐ Less than 10% of new players quit during the first 60 seconds.
- ☐ No single cause of death above 30% of all deaths (if one is, it is unfair or unclear).
- ☐ Crash-free sessions 99.5% or higher.

Do not build yet (after launch, only if the core is working)

- Multiplayer, story mode, open world, car customization beyond paint, extra currencies.